

HDOC – Experiment 8

Question 1

Write a menu-driven C program to input two numbers and swap them using a third variable and without using a third variable. Display the values before and after swapping.

❖ Objective

A **menu-driven** C program that allows the user to choose between two methods of swapping two numbers:

Swapping using a third variable and by swapping without using a third variable.

Inputs

- First number (A)
- Second number (B)
- User's menu choice

Output

- Values before swapping
- Values after swappingValues before swapping
- Values after swapping

❖ Test case table

Test Case	Input (A, B, Choice)	Expected Output	Actual Output	Result
1	10, 20, 1	Before: 10 20 After: 20 10	Same	Pass
2	25, 15, 2	Before: 25 15 After: 15 25	Same	Pass
3	-8, 12, 1	Before: -8 12 After: 12 -8	Same	Pass

❖ Pseudocode

BEGIN

Display menu

1. Swap using third variable
2. Swap without third variable

Read user's choice

Input first number

Input second number

Display values before swapping

IF choice == 1

Temp = A

A = B

B = Temp

ELSE IF choice == 2

A = A + B

B = A - B

A = A - B

ELSE

Display Invalid Choice

Display values after swapping

END

❖ Algorithm

1. Start.
2. Display the menu with two swapping methods.
3. Read the user's choice.
4. Read the two numbers.
5. Display the numbers before swapping.
6. If the choice is 1, swap the numbers using a temporary variable.
7. Otherwise, if the choice is 2, swap the numbers without using a temporary variable.
8. If the entered choice is invalid, display an error message.

9. Display the numbers after swapping.

10. Stop.

❖ Flowchart

START

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Display Menu

|



Read Choice

|



Read First and Second Number

|



Display Values Before Swapping

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Is Choice == 1 ?

/ \

Yes No

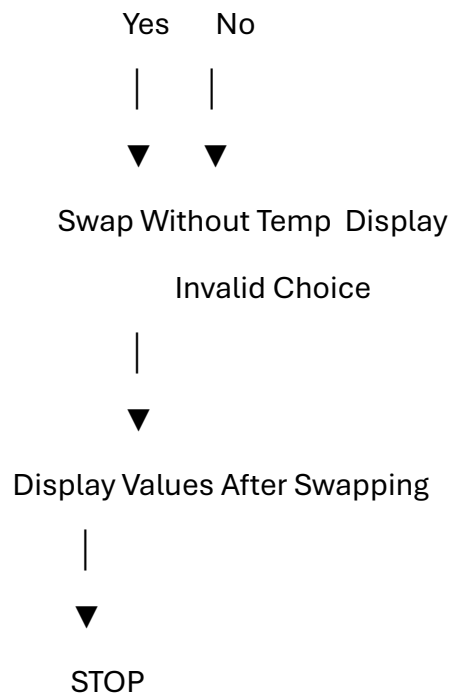
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Swap Using Temp Is Choice == 2 ?

/ \



❖ Program code in C

```
#include <stdio.h>

int main()
{
    int a, b, temp, choice;

    printf("===== MENU =====\n");
    printf("1. Swap using Third Variable\n");
    printf("2. Swap without Third Variable\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter First Number: ");
    scanf("%d", &a);
```

```
printf("Enter Second Number: ");  
scanf("%d", &b);
```

```
printf("\nBefore Swapping:\n");  
printf("First Number = %d\n", a);  
printf("Second Number = %d\n", b);
```

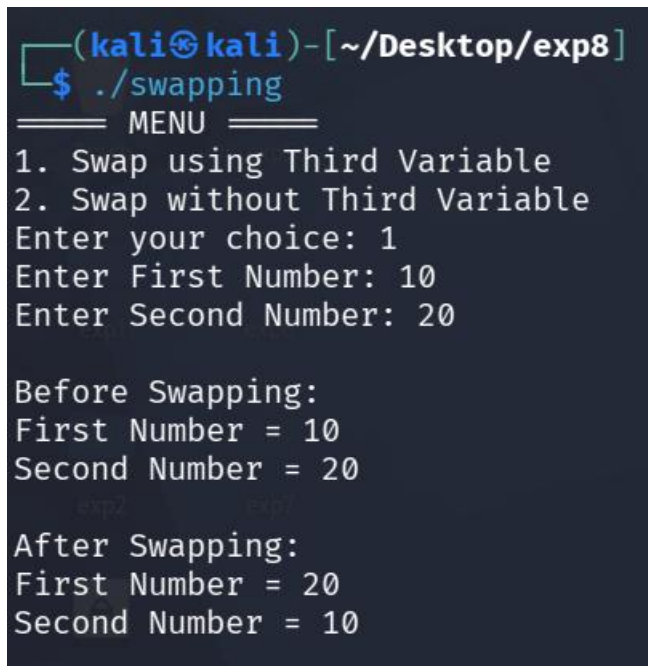
```
switch(choice)  
{  
    case 1:  
        temp = a;  
        a = b;  
        b = temp;  
        break;  
  
    case 2:  
        a = a + b;  
        b = a - b;  
        a = a - b;  
        break;  
  
    default:  
        printf("\nInvalid Choice!\n");  
        return 0;  
}
```

```
printf("\nAfter Swapping:\n");
```

```
printf("First Number = %d\n", a);  
printf("Second Number = %d\n", b);  
  
return 0;  
}
```

❖ Compile, Execute and Evaluate

Test case 1



```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./swapping  
===== MENU =====  
1. Swap using Third Variable  
2. Swap without Third Variable  
Enter your choice: 1  
Enter First Number: 10  
Enter Second Number: 20  
  
Before Swapping:  
First Number = 10  
Second Number = 20  
exp2 exp1  
After Swapping:  
First Number = 20  
Second Number = 10
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp8]
$ ./swapping
===== MENU =====
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 2
Enter First Number: 25
Enter Second Number: 15

Before Swapping:
First Number = 25
Second Number = 15

After Swapping:
First Number = 15
Second Number = 25
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp8]
$ ./swapping
===== MENU =====
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: -8
Enter Second Number: 12

Before Swapping:
First Number = -8
Second Number = 12

After Swapping:
First Number = 12
Second Number = -8
```

❖ Compilation and execution screenshot

```
(kali㉿kali)-[~/Desktop/exp8]
$ vi swapping.c

(kali㉿kali)-[~/Desktop/exp8]
$ gcc swapping.c -o swapping

(kali㉿kali)-[~/Desktop/exp8]
$ ./swapping
===== MENU =====
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: 10
Enter Second Number: 20

Before Swapping:
First Number = 10
Second Number = 20

After Swapping:
First Number = 20
Second Number = 10

(kali㉿kali)-[~/Desktop/exp8]
$ ./swapping
===== MENU =====
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 2
Enter First Number: 25
Enter Second Number: 15

Before Swapping:
First Number = 25
Second Number = 15
```

```
After Swapping:
First Number = 15
Second Number = 25

(kali㉿kali)-[~/Desktop/exp8]
$ ./swapping
===== MENU =====
1. Swap using Third Variable
2. Swap without Third Variable
Enter your choice: 1
Enter First Number: -8
Enter Second Number: 12

Before Swapping:
First Number = -8
Second Number = 12

After Swapping:
First Number = 12
Second Number = -8
```

Question 2

Write a menu-driven C program to find the sum of the first n natural numbers and the sum of squares of the first n natural numbers.

❖ Objective

A menu-driven C program that allows the user to choose between:

Finding the sum of the first n natural numbers. Finding the sum of the squares of the first n natural numbers.

The program should accept the value of n , perform the selected operation, and display the result.

Input

- Menu choice
- Value of n (positive integer)

Output

- Sum of the first n natural numbers **or**
- Sum of the squares of the first n natural numbers

Formula

Sum of First n Natural Numbers:

$$S = n(n+1)/2$$

Sum of Squares of First n Natural Numbers:

$$S = n(n+1)(2n+1)/6$$

❖ Test case table

Test Case	Input (n, Choice)	Expected Output	Actual Output	Result
1	10, 1	Sum = 55	Same	Pass
2	5, 2	Sum of Squares = 55	Same	Pass
3	20, 1	Sum = 210	Same	Pass

❖ Pseudocode

BEGIN

Display Menu

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers

Read choice

Read n

IF choice = 1 THEN

$\text{sum} = n \times (n + 1) / 2$

 Display sum

ELSE IF choice = 2 THEN

$\text{sum} = n \times (n + 1) \times (2 \times n + 1) / 6$

Display sum

ELSE

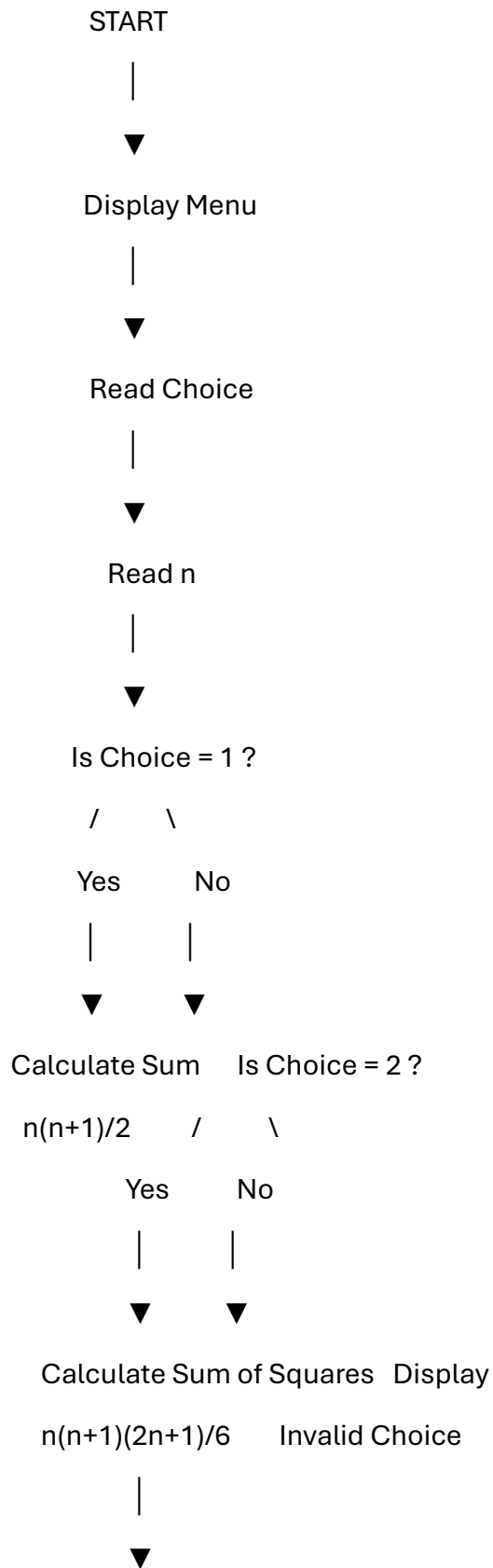
Display "Invalid Choice"

END

❖ Algorithm

1. Start.
2. Display the menu with two options.
3. Read the user's choice.
4. Read the value of n.
5. If the choice is 1, calculate the sum of the first n natural numbers.
6. If the choice is 2, calculate the sum of the squares of the first n natural numbers.
7. If the entered choice is invalid, display an error message.
8. Display the calculated result.
9. Stop.

❖ Flowchart



Display Result

|



STOP

❖ Program code in C

```
#include <stdio.h>

int main()
{
    int choice;
    long int n, sum;

    printf("==== MENU =====\n");
    printf("1. Sum of First n Natural Numbers\n");
    printf("2. Sum of Squares of First n Natural Numbers\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter the value of n: ");
    scanf("%ld", &n);

    switch(choice)
    {
        case 1:
            sum = n * (n + 1) / 2;
            printf("\nSum of first %ld natural numbers = %ld\n", n, sum);
```

```

        break;

    case 2:

        sum = n * (n + 1) * (2 * n + 1) / 6;

        printf("\nSum of squares of first %ld natural numbers = %ld\n", n, sum);

        break;

    default:

        printf("\nInvalid Choice!\n");

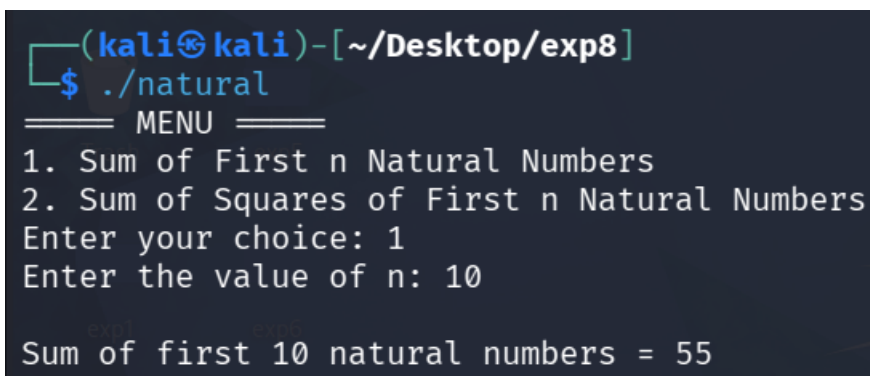
}

return 0;
}

```

❖ Compile, Execute and Evaluate

Test case 1



```

(kali㉿kali)-[~/Desktop/exp8]
$ ./natural
===== MENU =====
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 1
Enter the value of n: 10

Sum of first 10 natural numbers = 55

```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./natural  
===== MENU =====  
1. Sum of First n Natural Numbers  
2. Sum of Squares of First n Natural Numbers  
Enter your choice: 2  
Enter the value of n: 5  
  
Sum of squares of first 5 natural numbers = 55
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./natural  
===== MENU =====  
1. Sum of First n Natural Numbers  
2. Sum of Squares of First n Natural Numbers  
Enter your choice: 1  
Enter the value of n: 20  
  
Sum of first 20 natural numbers = 210
```

❖ Compilation and execution screenshot

```
(kali㉿kali)-[~/Desktop/exp8]
$ vi natural.c

(kali㉿kali)-[~/Desktop/exp8]
$ gcc natural.c -o natural

(kali㉿kali)-[~/Desktop/exp8]
$ ./natural
===== MENU =====
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 1
Enter the value of n: 10

Sum of first 10 natural numbers = 55

(kali㉿kali)-[~/Desktop/exp8]
$ ./natural
===== MENU =====
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 2
Enter the value of n: 5

Sum of squares of first 5 natural numbers = 55

(kali㉿kali)-[~/Desktop/exp8]
$ ./natural
===== MENU =====
1. Sum of First n Natural Numbers
2. Sum of Squares of First n Natural Numbers
Enter your choice: 1
Enter the value of n: 20

Sum of first 20 natural numbers = 210
```

Question 3

Write a menu-driven C program to calculate Simple Interest, Compound Amount, Compound Interest, and Total Payable Amount for given Principal, Rate, and Time.

❖ Objective

Develop a menu-driven C program that allows the user to choose one of the following operations:

Calculate Simple Interest (SI); Calculate Compound Amount (A); Calculate Compound Interest (CI); Calculate Total Payable Amount

The program should accept the principal (P), Rate of Interest (R), and Time (T) as input and display the required result based on the selected option.

Input

- Principal Amount (P)
- Rate of Interest (R) in percentage
- Time (T) in years
- Menu Choice

Output

- Simple Interest
- Compound Amount
- Compound Interest
- Total Payable Amount

Formula

Simple Interest:

$$SI = P \times R \times T / 100$$

Compound Amount:

$$A = P(1 + R/100)^T$$

Compound Interest:

$$CI = A - P$$

Total Payable Amount:

$$\text{Total Payable} = P + SI$$

❖ Test case table

Test Case	Input (P,R,T,Choice)	Expected Output	Actual Output	Result
1	10000, 10, 2, 1	SI = ₹2000.00	SAME	Pass
2	10000, 10, 2, 2	Amount = ₹12100.00	SAME	Pass
3	10000, 10, 2, 3	CI = ₹2100.00	SAME	Pass
4	10000, 10, 2, 4	Total Payable = ₹12000.00	SAME	Pass

❖ Pseudocode

BEGIN

Display Menu

1. Simple Interest

2. Compound Amount

3. Compound Interest

4. Total Payable Amount

Read Choice

Read Principal, Rate and Time

Calculate

$$SI = (P \times R \times T) / 100$$

$$\text{Amount} = P \times (1 + R/100)^T$$

$$CI = \text{Amount} - P$$

$$\text{Total} = P + SI$$

IF Choice = 1

Display SI

ELSE IF Choice = 2

Display Amount

ELSE IF Choice = 3

Display CI

ELSE IF Choice = 4

Display Total

ELSE

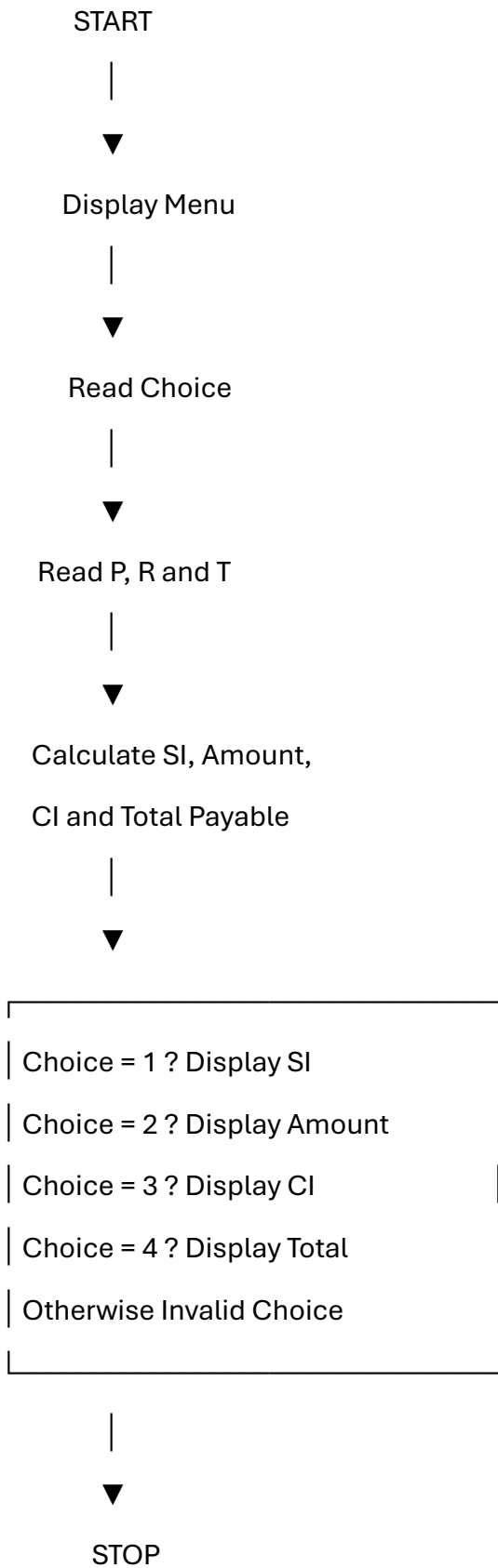
Display Invalid Choice

END

❖ Algorithm

1. Start.
2. Display the menu with four options.
3. Read the user's choice.
4. Read the principal amount, rate of interest, and time.
5. Calculate Simple Interest.
6. Calculate Compound Amount.
7. Calculate Compound Interest.
8. Calculate Total Payable Amount.
9. Display the result according to the selected menu option.
10. If the choice is invalid, display an error message.
11. Stop.

❖ Flowchart



❖ Program code in C

```
#include <stdio.h>

#include <math.h>

int main()
{
    int choice;

    double principal, rate, time;

    double simpleInterest, compoundAmount;

    double compoundInterest, totalPayable;


    printf("==== MENU =====\n");
    printf("1. Simple Interest\n");
    printf("2. Compound Amount\n");
    printf("3. Compound Interest\n");
    printf("4. Total Payable Amount\n");


    printf("Enter your choice: ");
    scanf("%d", &choice);


    printf("Enter Principal Amount: ");
    scanf("%lf", &principal);


    printf("Enter Rate of Interest: ");
    scanf("%lf", &rate);


    printf("Enter Time (Years): ");
    scanf("%lf", &time);
```

```
simpleInterest = (principal * rate * time) / 100.0;
compoundAmount = principal * pow((1 + rate / 100), time);
compoundInterest = compoundAmount - principal;
totalPayable = principal + simpleInterest;

switch(choice)
{
    case 1:
        printf("\nSimple Interest = Rs. %.2lf\n", simpleInterest);
        break;

    case 2:
        printf("\nCompound Amount = Rs. %.2lf\n", compoundAmount);
        break;

    case 3:
        printf("\nCompound Interest = Rs. %.2lf\n", compoundInterest);
        break;

    case 4:
        printf("\nTotal Payable Amount = Rs. %.2lf\n", totalPayable);
        break;

    default:
        printf("\nInvalid Choice!\n");
}
```

```
return 0;  
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./interest  
===== MENU =====  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 1  
Enter Principal Amount: 10000  
Enter Rate of Interest: 10  
Enter Time (Years): 2  
  
Simple Interest = Rs. 2000.00
```

Test case 2

```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./interest  
===== MENU =====  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 2  
Enter Principal Amount: 10000  
Enter Rate of Interest: 10  
Enter Time (Years): 2  
  
Compound Amount = Rs. 12100.00
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./interest  
===== MENU =====  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 3  
Enter Principal Amount: 10000  
Enter Rate of Interest: 10  
Enter Time (Years): 2  
  
Compound Interest = Rs. 2100.00
```

Test case 4

```
(kali㉿kali)-[~/Desktop/exp8]  
$ ./interest  
===== MENU =====  
1. Simple Interest  
2. Compound Amount  
3. Compound Interest  
4. Total Payable Amount  
Enter your choice: 4  
Enter Principal Amount: 10000  
Enter Rate of Interest: 10  
Enter Time (Years): 2  
  
Total Payable Amount = Rs. 12000.00
```

❖ Compilation and execution screenshot

```
(kali㉿kali)-[~/Desktop/exp8]
$ vi interest.c

(kali㉿kali)-[~/Desktop/exp8]
$ gcc interest.c -o interest -lm

(kali㉿kali)-[~/Desktop/exp8]
$ ./interest
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 1
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Simple Interest = Rs. 2000.00

(kali㉿kali)-[~/Desktop/exp8]
$ ./interest
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 2
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Compound Amount = Rs. 12100.00
```

```
(kali㉿kali)-[~/Desktop/exp8]
$ ./interest
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 3
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Compound Interest = Rs. 2100.00

(kali㉿kali)-[~/Desktop/exp8]
$ ./interest
===== MENU =====
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 4
Enter Principal Amount: 10000
Enter Rate of Interest: 10
Enter Time (Years): 2

Total Payable Amount = Rs. 12000.00
```

